Second Section.

Theta Functions.

§ 15. Presuppositions from function theory.

The concept of doubly periodic functions can be rightly grasped only when they are considered as functions of a complex argument

$$u = v + iw,$$

where i has, as usual, the meaning of $\sqrt{-1}$. The variable u is to us an independent and unboundedly variable magnitude, which, following Gauss, is represented geometrically by the points of a plane in such a way that the point whose coordinates in a rectangular system are v, w is regarded as the carrier of the value $u = v + iw$, and is briefly called the point u.

We restrict our consideration here to single-valued analytic functions $\varphi(u)$ which in the finite part have no essentially singular places. What happens at infinity we leave undecided. Thus we assume the following about the functions considered.

1. In every finite piece of surface there lies a finite number of points at which the function $\varphi(u)$ becomes infinite; these points are called points of discontinuity.

The number of the points of discontinuity in all, that is, in the whole infinite plane, can of course also be infinitely large.

2. If u_0 is a point not belonging to the points of discontinuity, then the function is developable into a series proceeding according to whole ascending positive powers of $u - u_0$, which is convergent in a circle having u_0 as center and reaching to the nearest point of discontinuity. One says that the function has, in the neighborhood of the point u_0, the character of an entire function.

3. If u_0 is a point of discontinuity, then there is a whole positive number m such that $(u-u_0)^m\varphi(u)$ remains finite, continuous, and different from zero at the point u_0, and hence is developable according to whole positive powers of $u - u_0$. In this case the point of discontinuity is called one of the mth order. From this there results an expansion of $\varphi(u)$ according to rising powers of $u - u_0$, which begins with $(u - u_0)^{-m}$ and converges in a circle described about u_0 and reaching to the nearest point of discontinuity. The coefficient of $(u - u_0)^{-1}$ in this expansion is called the residue of this function for the point u_0.

4. Cauchy's theorem. The integral

$$\frac{1}{2\pi i}\int \varphi(u)\,du,$$

extended in the positive sense over the boundary of a finite piece of surface which contains no point of discontinuity on its boundary line, is equal to the sum of the residues for the points of discontinuity lying in the interior of the piece of surface. As positive direction of integration one is here to regard the direction which lies with respect to the interior of the piece of surface as the v-axis does to the w-axis, so that, with the usual way of determining these axes, when the boundary is traversed positively the interior of the surface remains on the left.

5. The function

$$\psi(u) = \frac{d\log\varphi(u)}{du} = \frac{1}{\varphi(u)}\frac{d\varphi(u)}{du}$$

has only points of discontinuity of the first order, and if at a point u_0 the product $(u - u_0)^{-m}\varphi(u)$ is finite and different from zero, then m is the residue of $\psi(u)$ for this point. If m is positive, then u_0 is called a zero of mth order of $\varphi(u)$.

Accordingly the following is an immediate consequence of Cauchy's theorem.

6. The integral

$$\frac{1}{2\pi i}\int d\log\varphi(u),$$

extended in the positive direction over the boundary of a piece of surface, is equal to the number of zeros, diminished by the number of points of discontinuity, which lie in the interior of this piece of surface, zeros and points of discontinuity of mth order being counted as m such points of first order. Hence it follows that in the interior of a finite piece of surface there also lies only a finite number of zeros.

In the same sense one obtains:

7. The integral

$$\frac{1}{2\pi i}\int u\, d\log\varphi(u)$$

is equal to the sum of the values of u for which $\varphi(u)$ vanishes, diminished by the sum of the values of u for which $\varphi(u)$ becomes infinite.

8. A function which has no points of discontinuity at all in the finite part is called an entire function. An entire function which also remains finite at infinity is a constant. An entire function which also at infinity has no essentially singular place, so that there is therefore an exponent m such that $u^{-m}\varphi(u)$ does not become infinite for $u=\infty$, is a rational function.

9. An entire function whose reciprocal is likewise entire is called a unit function. Such a function becomes neither infinite nor zero in the finite part. Its logarithm is therefore likewise an entire function, and it follows that every unit function can be represented in the form $e^{g(u)}$, where $g(u)$ is an entire function. Conversely, every expression of this form is a unit function.

10. According to the fundamental investigations of Weierstrass on single-valued analytic functions, there is always an entire function $G(u)$ which vanishes at the points of discontinuity of a function $\varphi(u)$ and only at these, and this function $G(u)$ is determined by the points of discontinuity of $\varphi(u)$ up to a unit function as factor. Then $\varphi(u)G(u)=G_1(u)$ is likewise an entire function, and thus every function $\varphi(u)$ can be represented as the quotient of two entire functions

$$\varphi(u)=\frac{G_1(u)}{G(u)},$$

where $G(u)$ and $G_1(u)$ have no common zeros.

11. Numerator and denominator of this representation are uniquely determined by $\varphi(u)$ itself, apart from unit factors.

The functions $G(u)$, $G_1(u)$ can be decomposed into prime factors, that is, into factors which become zero at only one point. If one pursues this path for the doubly periodic functions, one arrives at the Weierstrass σ-functions. We shall, however, take another path, on which the ϑ-functions first meet us at a later place.

§ 16. Periodicity.

A function $\varphi(u)$ of u which has the property that, for a constant ω and for every value of u,

$$\varphi(u+\omega)=\varphi(u) \tag{1}$$

is called periodic, and ω is called the period.

The object of our investigation is functions $\varphi(u)$ with two periods ω_1,ω_2, of which we shall assume once for all that their ratio $\omega_2:\omega_1$ is not real and that the imaginary part of this ratio is positive. In the latter assumption there lies only a convention concerning the designation ω_1,ω_2. For if $\omega_2:\omega_1$ has a negative imaginary part, then that of $\omega_1:\omega_2$ is positive.

Every combination $m_1\omega_1+m_2\omega_2$ is likewise a period, when m_1,m_2 are integers.

In the u-plane we take an arbitrary point u_0 and denote the four points

$$u_0,\qquad u_0+\omega_1,\qquad u_0+\omega_1+\omega_2,\qquad u_0+\omega_2.$$

If we connect these four points in order by straight lines, returning from the last to the first, we obtain a parallelogram, which is called the period parallelogram. The straight sides of the period parallelogram may also be replaced by curved arcs, provided that opposite sides come to coincide by parallel displacement and the individual arcs form no loops.

If $\omega_1 = \alpha_1 + \beta_1 i$, $\omega_2 = \alpha_2 + \beta_2 i$ and $\alpha_1, \alpha_2, \beta_1, \beta_2$ are real, then by the assumption $\alpha_1\beta_2 - \alpha_2\beta_1$ is positive, and the geometrical meaning of this magnitude is the area of the period parallelogram.

By adjoining congruent period parallelograms the whole u-plane can be covered simply and without gaps. Corresponding points of two different such parallelograms are representatives of u-values which differ by integral multiples of the periods, and which are called congruent modulo (ω_1, ω_2), or, if no doubt is possible, simply congruent. The sign of congruence is

$$u \equiv u' \pmod{\omega_1, \omega_2},$$

and means that u' has the form $u + m_1\omega_1 + m_2\omega_2$, where m_1, m_2 are integers.

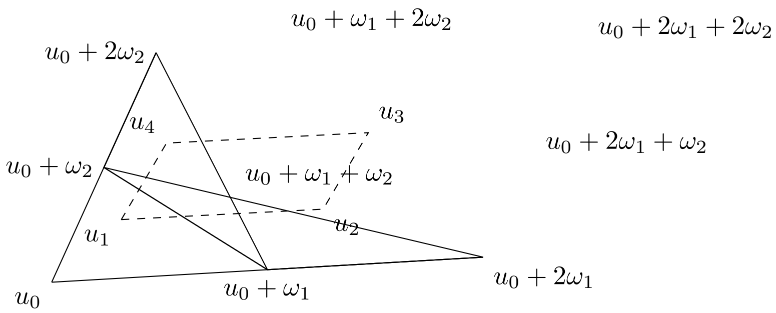

Figure 1: Fig. 1.

Thus Fig. 1 represents four period parallelograms laid alongside one another; the points u_1, u_2, u_3, u_4 are congruent:

$$u_2 = u_1 + \omega_1, \qquad u_3 = u_1 + \omega_1 + \omega_2, \qquad u_4 = u_1 + \omega_2,$$

and u_1, u_2, u_3, u_4 likewise form the corners of a period parallelogram.

The doubly periodic function $\varphi(u)$ has the same value at all congruent points, and the entire stock of values of this function is therefore exhausted in one period parallelogram, if of two opposite sides only one is counted with the parallelogram.

From our general assumptions we draw the conclusion:

1. There exists, apart from the constant, no doubly periodic function which is free of points of discontinuity in the period parallelogram; for such a function would be finite in the whole u-plane and hence by § 15, 8 would have to be a constant.

If $\varphi(u)$ is a doubly periodic function with periods ω_1, ω_2, then the integral taken over the boundary of the period parallelogram is, because of (1),

$$\int d\log\varphi(u) = \int_{u_0}^{u_0+\omega_1} [d\log\varphi(u) - d\log\varphi(u+\omega_2)] - \int_{u_0}^{u_0+\omega_2} [d\log\varphi(u) - d\log\varphi(u+\omega_1)] = 0.$$

From this follows, by § 15, 6, the theorem:

2. A doubly periodic function becomes zero at just as many points in a period parallelogram as it becomes infinite. If m is this number, then m is called the order of the doubly periodic function. By 1. there are no doubly periodic functions of order zero.

If in the proof one replaces the function $\varphi(u)$ by $\varphi(u) - c$, it follows that a doubly periodic function of mth order assumes every arbitrary value c at equally many points.

If we take the integral of the formula in 7., § 15 over the boundary of the period parallelogram, we obtain

$$\begin{aligned}\int u\,d\log\varphi(u) &= \int_{u_0}^{u_0+\omega_1}[u\,d\log\varphi(u) - (u+\omega_2)d\log\varphi(u+\omega_2)]\\ &\quad - \int_{u_0}^{u_0+\omega_2}[u\,d\log\varphi(u) - (u+\omega_1)d\log\varphi(u+\omega_1)]\\ &= -\omega_2\int d\log\varphi(u) + \omega_1\int d\log\varphi(u).\end{aligned}$$

Now, because of the many-valuedness of the logarithm,

$$\int d\log\varphi(u) = \log\varphi(u+\omega_1) - \log\varphi(u) = 2\pi i m_1,$$

$$\int d\log\varphi(u) = \log\varphi(u+\omega_2) - \log\varphi(u) = -2\pi i m_2,$$

where m_1 and m_2 are undetermined integers.

If therefore the function $\varphi(u)$ becomes infinite at the points $\alpha_1, \alpha_2, \dots, \alpha_m$ and zero at the points $\beta_1, \beta_2, \dots, \beta_m$, then, by § 15, 7,

$$\sum\alpha \equiv \sum\beta \pmod{\omega_1, \omega_2}.$$

If one replaces $\varphi(u)$ by $\varphi(u) - c$, where c denotes an arbitrary value, the important theorem follows:

3. The sum of the argument values at which a doubly periodic function assumes one and the same value c in the interior of a period parallelogram is independent of c, or changes under continuous variation of c at most by jumps of a period.

4. From this it follows also that there is no doubly periodic function of the first order.

For such a function would have to assume every value c at one point u. If it were therefore equal to c' at a point u' and equal to c' at a different point u'', then by 3. $u' = u''$ would follow, contrary to the assumption.

By a doubly periodic function of the second kind one understands, following Hermite, a function $\psi(u)$ which, with respect to the periods ω_1, ω_2, behaves according to the equations

$$\psi(u+\omega_1) = A_1\psi(u), \qquad \psi(u+\omega_2) = A_2\psi(u),$$

where A_1, A_2 are constants.

5. Also for a doubly periodic function of the second kind the theorem holds that it becomes zero and infinite at equally many points of the period parallelogram. If the number of these points is m, then here too the function is called of the mth order.

The proof is the same as that of theorem 2. But here the extension no longer holds that the function $\psi(u)$ assumes every value equally often, because $\psi(u) - c$ is no longer periodic.

6. A doubly periodic function $\psi(u)$ of the second kind and of order zero is necessarily a constant or an exponential function. For if $\psi(u)$ does not become zero, then

$$\varphi(u) = \frac{\psi'(u)}{\psi(u)}$$

is a doubly periodic function of the first kind which does not become infinite, and hence by 1. is a constant a. It follows that

$$\psi(u) = Ae^{au},$$

where A is a new constant.

§ 17. The functions T.

If doubly periodic functions $\varphi(u)$ of the mth order exist, then they must be fractional functions. We therefore set them, according to § 15, 10, in the form

$$\varphi(u) = \frac{G_1(u)}{G(u)}. \tag{1}$$

If $G_1(u)$ and $G(u)$ are entire functions without a common zero, then

$$\frac{G(u+\omega_1)}{G(u)} = \frac{G_1(u+\omega_1)}{G_1(u)}, \qquad \frac{G(u+\omega_2)}{G(u)} = \frac{G_1(u+\omega_2)}{G_1(u)}.$$

In these equations the three functions $G(u)$, $G(u+\omega_1)$, $G(u+\omega_2)$ have the same m zeros, namely the points of discontinuity of $\varphi(u)$. Their quotients are therefore unit functions, and by § 15, 9, if $l_1(u), l_2(u)$ are entire functions,

$$G(u+\omega_1) = e^{l_1(u)}G(u), \qquad G(u+\omega_2) = e^{l_2(u)}G(u), \tag{2}$$

and the function $G_1(u)$ satisfies the same equations.

If

$$\varphi(u) = \frac{T_1(u)}{T(u)} \tag{3}$$

is a second representation of $\varphi(u)$ of the form (1), then

$$T(u) = e^{g(u)}G(u), \tag{4}$$

where $g(u)$ is again an entire function, and if therefore, by (2),

$$T(u+\omega_1) = e^{l'_1(u)}T(u), \qquad T(u+\omega_2) = e^{l'_2(u)}T(u), \tag{5}$$

then

$$l'_1(u) = l_1(u) + g(u) - g(u+\omega_1), \qquad l'_2(u) = l_2(u) + g(u) - g(u+\omega_2).$$

Everything therefore depends on being able, under some suitable assumption about the entire functions $l_1(u), l_2(u)$, to form entire functions $T(u)$ which satisfy the conditions (5) and vanish at m arbitrarily given points of the period parallelogram. By the quotients of two such functions with the same $l_1(u), l_2(u)$ we can then represent all doubly periodic functions $\varphi(u)$, and the more general functions $G(u)$ which accomplish the same thing are obtained by (4) through adjoining an arbitrary unit factor.

If we wanted to assume $l_1(u), l_2(u)$ to be constants, then $T(u)$ would be an entire doubly periodic function of the second kind and consequently by § 16, 6 an exponential function, from which no doubly periodic functions can be formed. The simplest assumption that remains to us after this is that $l_1(u), l_2(u)$ are linear functions of u, including as a special case also the case that one of the two functions l_1, l_2 is constant.

Thus we set up the following definition:

1. A T-function, $T(u)$, is an entire function of u which satisfies the following two equations:

$$\begin{aligned} T(u+\omega_1) &= e^{-\pi i[a_1(2u+\omega_1)+b_1]}T(u), \\ T(u+\omega_2) &= e^{-\pi i[a_2(2u+\omega_2)+b_2]}T(u). \end{aligned} \tag{6}$$

Here a_1, b_1, a_2, b_2 are constants, that is, magnitudes independent of u. The periods ω_1, ω_2 are taken here throughout as constants given once for all, whose ratio $\omega_2 : \omega_1$ has a positive imaginary part.

The factors

$$e_1 = e_1(u) = e^{-\pi i[a_1(2u+\omega_1)+b_1]}, \qquad e_2 = e_2(u) = e^{-\pi i[a_2(2u+\omega_2)+b_2]}$$

are called the two periodicity factors of the function T.

The periodicity factors do not change if b_1, b_2 are changed by even integers. That functions of this kind really exist will first appear in the further course of the investigation.

2. If the function $T(u)$ vanishes at any point, then it also vanishes at all congruent points. If it vanishes at m points of the period parallelogram, then it is called of the mth order.

From this definition it follows at once:

3. By multiplying two T-functions T, T' of orders m, m' one obtains a T-function of order $m+m'$. The periodicity factors of the product TT' are the products of the periodicity factors of T and T'.

We first ask about the possibility of T-functions of zeroth order. If $L(u)$ is such a function and $L'(u)$ its derivative, then by twice differentiating (6) it follows that

$$\frac{dL(u+\omega_1)}{du\,L(u+\omega_1)} = \frac{dL(u)}{du\,L(u)},$$

and correspondingly from the second equation (6). Hence $d^2 \log L(u)/du^2$, as an entire doubly periodic function, is a constant, and consequently $\log L(u)$ is an entire function of the second degree in u. We therefore obtain:

4. A T-function of zeroth order is of the form

$$L(u) = Ce^{-\pi i(\lambda u^2 + \mu u)}, \tag{7}$$

where λ, μ, C are constants.

That every function of this form is a T-function of zeroth order is seen immediately.

The product

$$T'(u) = L(u)T(u) \tag{8}$$

is, like $T(u)$, a T-function of mth order and vanishes at the same points.

The periodicity factors of T' are obtained from those of T when the exponents a_1, a_2, b_1, b_2 are replaced by

$$a_1' = a_1 + \lambda\omega_1, \quad b_1' = b_1 + \mu\omega_1, \qquad a_2' = a_2 + \lambda\omega_2, \quad b_2' = b_2 + \mu\omega_2, \tag{9}$$

where however the b_1', b_2' are determined only up to additive even integers.

5. By (9) one can determine μ always, and in only one way, so that b_1', b_2' become real.

For if one sets the imaginary parts of b_1', b_2' equal to zero, one obtains for the imaginary and real part of μ two linear equations whose determinant, by § 16, is the area of the period parallelogram and therefore different from zero.

The periodicity factors and the zeros of a T-function of mth order stand in a certain dependence on one another, which we obtain easily from the theorems of § 15. First the order m is obtained when one extends the integral

$$\int d\log T(u) = 2\pi i\, m$$

over the boundary of the period parallelogram. For simplicity let the corner u_0 lie at the origin of coordinates; then this integral can be decomposed as

$$2\pi i m = \int_0^{\omega_1} d[\log T(u) - \log T(u+\omega_2)] - \int_0^{\omega_2} d[\log T(u) - \log T(u+\omega_1)],$$

and by (6)

$$d[\log T(u) - \log T(u+\omega_2)] = 2\pi i a_2, \qquad d[\log T(u) - \log T(u+\omega_1)] = 2\pi i a_1.$$

It follows that

$$a_2\omega_1 - a_1\omega_2 = m. \tag{10}$$

If we further denote by $\alpha_1, \alpha_2, \dots, \alpha_m$ the zeros of a T-function in the interior of a period parallelogram, then from § 15, 7 we obtain

$$2\pi i \sum \alpha = \int u \, d \log T(u),$$

when the integral is again extended over the boundary of the parallelogram. But

$$\begin{aligned}\int u \, d \log T(u) = & \int_0^{\omega_1} [u \, d \log T(u) - (u + \omega_2) d \log T(u + \omega_2)] \\ & - \int_0^{\omega_2} [u \, d \log T(u) - (u + \omega_1) d \log T(u + \omega_1)] \, .\end{aligned}$$

The first of these two integrals is, by (6),

$$\begin{aligned}& -\omega_2 \int_0^{\omega_1} d \log T(u) + 2\pi i a_2 \int_0^{\omega_1} (u + \omega_2) \, du \\ & = -\omega_2 [\log T(\omega_1) - \log T(0)] + \pi i a_2 \omega_1 (\omega_1 + 2\omega_2) \\ & = \pi i \omega_2 (a_1 \omega_1 + b_1) + \pi i a_2 (\omega_1^2 + 2\omega_1 \omega_2) + 2\pi i N_2 \omega_2,\end{aligned}$$

where N_2, because of the many-valuedness of the logarithm, is an integer not more closely determined. Likewise, for the second of the integrals one obtains

$$-\pi i \omega_1 (a_2 \omega_2 + b_2) - \pi i a_1 (\omega_2^2 + 2\omega_1 \omega_2) + 2\pi i N_1 \omega_1,$$

and consequently, written as a congruence,

$$\sum \alpha \equiv \frac{1}{2}(b_1 \omega_2 - b_2 \omega_1) + \frac{m}{2}(\omega_1 + \omega_2). \tag{11}$$

This sum, or rather the totality of the numbers congruent with it modulo (ω_1, ω_2), is called the character of the T-function.

The character of the function T is not changed if one replaces T by one of the functions LT. Thus by 5. one can also represent the character in the form

$$\sum \alpha \equiv \frac{1}{2}(g_1 \omega_2 - g_2 \omega_1) + \frac{m}{2}(\omega_1 + \omega_2), \tag{12}$$

where g_1, g_2 are real. These real numbers are determined by the character up to multiples of 2 and can take every value between 0 and 2. The symbol (g_1, g_2) is called the characteristic of the T-function. From the characteristic the character of the T-function is determined by (12).

If γ is the character of a function $T(u)$ of order m and v is an arbitrary constant, then $T(u+v)$ has the character

$$\gamma' = \gamma - vm.$$

Accordingly, when $m > 0$, the different characters can be reduced to one another.

With the help of equation (10) one can combine the two equations (6) into a general one. If in the first equation (6) one repeatedly changes u into $u + \omega_1$, one obtains the system

$$T(u + \omega_1) = e^{-\pi i [a_1 (2u + \omega_1) + b_1]} T(u),$$

$$T(u + 2\omega_1) = e^{-\pi i [a_1 (2u + 3\omega_1) + b_1]} T(u + \omega_1),$$
$$T(u + n_1 \omega_1) = e^{-\pi i [a_1 (2u + (2n_1 - 1)\omega_1) + b_1]} T(u + (n_1 - 1)\omega_1),$$

and by multiplication of all these formulae:

$$T(u + n_1 \omega_1) = e^{-\pi i [a_1 (2n_1 u + n_1^2 \omega_1) + n_1 b_1]} T(u). \tag{13}$$

This formula is first derived for a positive integral n_1. But if in it one replaces u by $u - n_1\omega_1$, its correctness for negative n_1 also results.

Likewise one obtains

$$T(u + n_2\omega_2) = e^{-\pi i[a_2(2n_2u+n_2^2\omega_2)+n_2b_2]}T(u), \tag{14}$$

and if in this formula one changes u into $u + n_1\omega_1$ and again applies (13),

$$T(u + n_1\omega_1 + n_2\omega_2) = e^{-\pi i[2(a_1n_1+a_2n_2)u+2a_2n_1n_2\omega_1+a_1n_1^2\omega_1+a_2n_2^2\omega_2+n_1b_1+n_2b_2]}T(u). \tag{15}$$

But by (10)

$$2a_2n_1n_2\omega_1 = a_1n_1n_2\omega_1 + a_2n_1n_2\omega_2 + mn_1n_2,$$

and therefore this last formula becomes

$$T(u + n_1\omega_1 + n_2\omega_2) = e^{-\pi i[(n_1a_1+n_2a_2)(2u+n_1\omega_1+n_2\omega_2)+b_1n_1+b_2n_2+mn_1n_2]}T(u). \tag{16}$$

Here n_1 and n_2 are arbitrary positive or negative integers.

If one multiplies two T-functions of order m and m' and of characters γ, γ', there arises a new T-function whose order is $m + m'$ and whose character is $\gamma + \gamma'$. If (g_1, g_2) and (g_1', g_2') are the characteristics of the two factors, then $(g_1 + g_1', g_2 + g_2')$ is the characteristic of the product.

§ 18. Relations between related T-functions.

Two T-functions with the same periods, the same order, and the same character shall be called related. If $T'(u)$ is a T-function related to $T(u)$, and if a_1', a_2', b_1', b_2' have the same meaning for T' as a_1, a_2, b_1, b_2 for T, then, in consequence of the assumed relatedness,

$$a_1'\omega_2 - a_2'\omega_1 = a_1\omega_2 - a_2\omega_1, \tag{1}$$
$$b_1'\omega_2 - b_2'\omega_1 = b_1\omega_2 - b_2\omega_1 + 2n_1\omega_2 - 2n_2\omega_1, \tag{2}$$

where n_1, n_2 are integers.

Therefore λ and μ can be determined according to § 17, (9) so that the periodicity factors of the function

$$e^{-\pi i(\lambda u^2+\mu u)}T(u)$$

become the same as those of the function $T'(u)$, and hence we have the theorem:

1. Related T-functions can, by adjoining a T-function of zeroth order

$$L = Ce^{-\pi i(\lambda u^2+\mu u)}$$

as factor, be given the same periodicity factors.

Furthermore, from the equality of the characters it follows immediately:

2. If related T-functions of mth order have $m - 1$ common zeros in the period parallelogram, then they also have the mth in common; and from this:

3. Between at most $m+1$ related T-functions of mth order $T, T_1, \ldots, T_m$ there exists an identical equation of the form

$$LT + L_1T_1 + L_2T_2 + \cdots + L_mT_m = 0.$$

For if this relation does not already hold for $L = 0$, then one can first determine the constants available in $L_1, L_2, \ldots, L_m$ so that all the products $L_1T_1, L_2T_2, \ldots, L_mT_m$ receive the same periodicity factors, and then the remaining constants so that $m - 1$, and consequently all, zeros of the function $L_1T_1 + L_2T_2 + \cdots + L_mT_m$ coincide with the zeros of the function T. From this, however, it follows that the ratio of $L_1T_1 + L_2T_2 + \cdots + L_mT_m$ to T is equal to a T-function of zeroth order, thus equal to $-L$, as was to be proved.

4. The quotients of related T-functions can be transformed into doubly periodic functions by adjoining a factor L.

5. If we assume the existence of a T-function of the first order $t(u)$, then every T-function of mth order can be derived from it.

Namely, let γ be the character of $t(u)$ and let $\alpha_1, \alpha_2, \ldots, \alpha_m$ be arbitrarily given values. Then the function

$$t(u - \alpha_i + \gamma) = t_i(u)$$

vanishes at the point α_i and all points congruent with α_i, but at no other.

The product

$$T(u) = t_1(u)t_2(u) \cdots t_m(u)$$

is therefore a T-function of mth order with the arbitrarily given zeros $\alpha_1, \alpha_2, \ldots, \alpha_m$.

6. From this it is easy to prove that every doubly periodic function which has only a finite number of points of discontinuity in a period parallelogram can be represented as the quotient of two T-functions.

Let $\varphi(u)$ be a doubly periodic function with periods ω_1, ω_2 and with points of discontinuity $\alpha_1, \alpha_2, \ldots, \alpha_m$, which may also partly coincide to points of discontinuity of higher order.

If according to 5. one determines a function $T(u)$ with zeros $\alpha_1, \alpha_2, \ldots, \alpha_m$, then

$$T(u)\varphi(u) = T_1(u)$$

is a T-function of mth order, and

$$\varphi(u) = \frac{T_1(u)}{T(u)}.$$

The functions $T(u), T_1(u)$ are related, since they have the same periodicity factors and hence also the same characteristic.

The zeros of $T_1(u)$ are at the same time the zeros of $\varphi(u)$, and the sum of these zero values is therefore congruent with the sum of the discontinuity values.

§ 19. T-functions of first order.

We now pass to the closer determination of the T-functions of first order, which we denote by $t(u)$, and from which, as we have seen, the remaining T-functions can be derived.

From 3. of the preceding paragraph it follows that two t-functions of the same characteristic differ only by a factor of the form

$$Ce^{-\pi i(\lambda u^2 + \mu u)},$$

and we shall now, by an extension of the definition, determine this factor still more closely.

This shall be done, as is possible without restriction of generality by § 17, (9), so that in the periodicity factors

$$a_1 = 0, \qquad b_1 = g_1, \qquad b_2 = g_2$$

when (g_1, g_2) is the characteristic; because of the relation

$$a_2\omega_1 - a_1\omega_2 = 1$$

one then has

$$a_2 = \frac{1}{\omega_1},$$

and the conditions for this t-function read:

$$t(u + \omega_1) = e^{-\pi i g_1} t(u), \qquad t(u + \omega_2) = e^{-\pi i \frac{2u + \omega_2}{\omega_1}} e^{-\pi i g_2} t(u). \tag{1}$$

By this the function $t(u)$ is determined up to a factor independent of u. In order to determine this factor still more closely, we consider also the dependence of the function t on ω_1, ω_2, and denote it, regarding g_1, g_2 as given numbers independent of ω_1, ω_2, by $t(u, \omega_1, \omega_2)$. It then appears that if h is an arbitrary factor,

$$t(hu, h\omega_1, h\omega_2)$$

likewise satisfies the conditions (1), and hence that

$$\frac{t(hu, h\omega_1, h\omega_2)}{t(u, \omega_1, \omega_2)} = f(\omega_1, \omega_2) \tag{2}$$

is independent of u.

But the function $f(\omega_1, \omega_2)$ can always be put into the form

$$f(\omega_1, \omega_2) = \frac{\varphi(h\omega_1, h\omega_2)}{\varphi(\omega_1, \omega_2)}.$$

One has only, if $t(u)$ does not vanish for $u = 0$, to put

$$\varphi(\omega_1, \omega_2) = t(0, \omega_1, \omega_2),$$

and if $t(0) = 0$, for instance

$$\varphi(\omega_1, \omega_2) = \omega_1 t'(0, \omega_1, \omega_2),$$

where t' denotes the derivative of t with respect to u. If therefore we now again denote

$$\frac{t(u, \omega_1, \omega_2)}{\varphi(\omega_1, \omega_2)} \tag{3}$$

by $t(u, \omega_1, \omega_2)$, then one can add to the conditions (1) the condition that for an arbitrary h

$$t(u, \omega_1, \omega_2) = t(hu, h\omega_1, h\omega_2). \tag{4}$$

The function t now depends only on two variables, namely the ratios $u : \omega_1 : \omega_2$. We put in (4)

$$h = \frac{1}{\omega_1}, \qquad \frac{u}{\omega_1} = v, \qquad \frac{\omega_2}{\omega_1} = \omega,$$

and obtain

$$t(u, \omega_1, \omega_2) = \vartheta\left(\frac{u}{\omega_1}, \frac{\omega_2}{\omega_1}\right) = \vartheta(v, \omega), \tag{5}$$

where ϑ is a new function sign.

Thus a new function $\vartheta(v)$ is defined, of only two variables v, ω, the first of which is unrestrictedly variable, while ω has, by the assumption made in § 16, a positive imaginary constituent; v is called the argument, ω the modulus of the ϑ-function.

§ 20. The ϑ-function.

The function $\vartheta(v)$ is an entire function of v which satisfies the conditions

$$\vartheta(v + 1) = e^{-\pi i g_1}\vartheta(v), \qquad \vartheta(v + \omega) = e^{-\pi i(2v+\omega)}e^{-\pi i g_2}\vartheta(v), \tag{1}$$

and is thereby determined up to a factor independent of v. It is therefore included among the t-functions as a special case, while conversely the general t-function can be expressed by the ϑ-function in the way

$$t(u) = Ce^{-\pi i(\lambda u^2+\mu u)}\vartheta\left(\frac{u}{\omega_1}, \frac{\omega_2}{\omega_1}\right),$$

where C, λ, μ are arbitrary magnitudes, constant or depending on ω_1, ω_2.

The function ϑ still depends on the numbers g_1, g_2 occurring in the characteristic, and if a notation for this dependence is necessary, one shall write for $\vartheta(v, \omega)$

$$\vartheta_{g_1,g_2}(v, \omega);$$

here g_1, g_2 may be arbitrary numbers. By what was said earlier it is enough if we assume them real and between 0 and 2.

Corresponding to the T-functions of higher order we shall also introduce Θ-functions of mth order, and understand by this an entire function of v which satisfies the conditions

$$\Theta(v + 1) = e^{-\pi i g_1}\Theta(v), \qquad \Theta(v + \omega) = e^{-m\pi i(2v+\omega)}e^{-\pi i g_2}\Theta(v). \tag{2}$$

Here also the characteristic g_1, g_2 can be included in the notation:

$$\Theta_{g_1,g_2}(v, \omega).$$

These functions are contained as special cases among the T-functions, and one can form general T-functions in the way

$$Ce^{-\pi i(\lambda u^2+\mu u)}\Theta\left(\frac{u}{\omega_1}, \frac{\omega_2}{\omega_1}\right). \tag{3}$$

Thus from § 18, 3 there results the fundamental theorem for the Θ-functions, which we state as follows.

If we call Θ-functions with the same periods, equal order, and equal characteristic related, and if furthermore we call a system of functions linearly dependent or independent according as a homogeneous linear relation with constant coefficients, not all vanishing, exists between these functions or not, then the theorem holds:

$m + 1$ related Θ-functions of mth order are always linearly dependent;

or:

from m linearly independent related Θ-functions of mth order

every other related Θ-function can be linearly composed, with constant coefficients.

This theorem is of the most fundamental importance for our following considerations; from it, by § 18, 3, it follows that every T-function of mth order can, by use of formula (3), be composed out of m linearly independent Θ-functions.

By § 17, (11), the sum of the m argument values for which the function $\Theta_{g_1,g_2}(v)$ vanishes in the period parallelogram is congruent modulo $(1, \omega)$ to

$$m\frac{1 + \omega}{2} + \frac{g_1\omega - g_2}{2}.$$

The definition of the function ϑ given up to now leaves undetermined a factor which may be a function of ω. But by an addition to the definition we can determine this factor more closely.

If the defining equations (1) are differentiated twice with respect to v and once with respect to ω, regarding g_1, g_2 as constants, then by a simple calculation it follows that the function

$$\frac{\partial^2\vartheta(v, \omega)}{\partial v^2} - 4\pi i\,\frac{\partial\vartheta(v, \omega)}{\partial\omega}$$

itself satisfies the conditions (1), and therefore has the form

$$\varphi(\omega)\vartheta(v, \omega),$$

where $\varphi(\omega)$ is constant with respect to v, hence a function of ω alone. If one now sets

$$e^{-\frac{1}{4\pi i}\int\varphi(\omega)\,d\omega}\vartheta(v,\omega)$$

equal to a new ϑ, then for this one obtains the partial differential equation

$$\frac{\partial^2\vartheta(v,\omega)}{\partial v^2} - 4\pi i\,\frac{\partial\vartheta(v,\omega)}{\partial\omega} = 0, \tag{4}$$

and by (1) and (4) the function ϑ is now defined up to a factor independent of v and ω, thus depending only on g_1, g_2.

§ 21. The theta functions of different characteristics. Main characteristics.

Before we go on to the determination of the remaining factor in the ϑ-functions that is independent of v and ω, we shall now reduce the different characteristics to one another, which by § 17 is always possible. If $\vartheta(v)$ denotes the ϑ-function belonging to the characteristic $(0,0)$, then

$$\Phi(v,\omega) = e^{-\pi i g_1 v}\,\vartheta\left(v - \frac{g_1\omega - g_2}{2}\right)$$

is a function satisfying the conditions (1) of § 20. But, with regard to the differential equation (4), one has

$$\frac{\partial^2\Phi}{\partial v^2} - 4\pi i\,\frac{\partial\Phi}{\partial\omega} = -\pi^2 g_1^2\Phi,$$

and if therefore we put

$$\vartheta_{g_1,g_2}(v) = e^{\frac{\pi i\omega g_1^2}{4} - \pi i g_1 v + \frac{g_1 g_2 \pi i}{2}}\,\vartheta\left(v - \frac{g_1\omega - g_2}{2}\right), \tag{1}$$

then the differential equation (4) of § 20 is satisfied by all these functions. Thus the functions ϑ_{g_1,g_2} are determined up to one numerical factor common to all.

From formula (1) a more general formula can be derived by replacing v by

$$v - \frac{g_1'\omega - g_2'}{2}.$$

One obtains

$$\vartheta_{g_1,g_2}\left(v - \frac{g_1'\omega - g_2'}{2}\right) = e^{-\frac{\pi i\omega}{4}g_1'^2 + \pi i g_1' v - \pi i g_1 g_2' - \frac{\pi i}{2}g_1'(g_2+g_2')}\vartheta_{g_1+g_1',g_2+g_2'}(v). \tag{2}$$

From (2) there also follows, using (1) of § 20, if one sets $g_1' = 2, g_2' = 0$, or $g_1' = 0, g_2' = 2$:

$$\vartheta_{g_1+2,g_2}(v) = e^{2\pi i g_2}\vartheta_{g_1,g_2}(v), \qquad \vartheta_{g_1,g_2+2}(v) = e^{\pi i g_1}\vartheta_{g_1,g_2}(v), \tag{3}$$

and by (2) at the same time the periodicity of the functions $\vartheta_{g_1,g_2}(v)$ is completely expressed. For example, if μ and ν are integers, one obtains

$$\vartheta_{00}\left(v - \frac{(2\nu-1)\omega + 2\mu - 1}{2}\right) = (-1)^{\nu+1} i\, e^{-\frac{\pi i\omega}{4}(2\nu-1)^2} e^{\pi i(2\nu-1)v}\vartheta_{11}(v), \tag{4}$$

$$\vartheta_{11}(v - \nu\omega - \mu) = (-1)^{\mu+\nu} e^{-\pi i\omega\nu^2} e^{2\pi i\nu v}\vartheta_{11}(v). \tag{5}$$

Formula (2) is valid for quite arbitrary, even complex, values of g_1, g_2.

If in the defining formulae of the ϑ-functions § 20, (1) one replaces v by $-v-1$ and by $-v-\omega$, and also replaces g_1, g_2 by $-g_1, -g_2$, then it is seen that $\vartheta_{-g_1,-g_2}(-v)$ satisfies the same conditions as $\vartheta_{g_1,g_2}(v)$, and hence that

$$\vartheta_{g_1,g_2}(v), \qquad \vartheta_{-g_1,-g_2}(-v)$$

are identical up to a constant factor; in particular, as follows from $v=0$,

$$\vartheta_{0,0}(v) = \vartheta_{0,0}(-v),$$

and then formula (1), when v, g_1, g_2 are replaced by $-v, -g_1, -g_2$, gives

$$\vartheta_{g_1,g_2}(v) = \vartheta_{-g_1,-g_2}(-v). \tag{6}$$

If the elements g_1, g_2 are integers, then (g_1, g_2) is called a main characteristic. There are four essentially different ones, namely

$$(0,0), \qquad (0,1), \qquad (1,0), \qquad (1,1),$$

and therefore also four essentially different main ϑ-functions:

$$\vartheta_{00}(v), \qquad \vartheta_{01}(v), \qquad \vartheta_{10}(v), \qquad \vartheta_{11}(v), \tag{7}$$

of which the first is also denoted by $\vartheta(v)$.

In what follows, by characteristics and ϑ-functions only main characteristics and main functions will be meant.

Formula (2), applied to this system of functions, gives the following frequently used table:

$$\begin{aligned}
\vartheta_{00}(v) &= \vartheta_{01}\left(v+\frac{1}{2}\right) = \varepsilon\,\vartheta_{10}\left(v+\frac{\omega}{2}\right) = \varepsilon\,\vartheta_{11}\left(v+\frac{1+\omega}{2}\right),\\
\vartheta_{01}(v) &= \vartheta_{00}\left(v+\frac{1}{2}\right) = -i\varepsilon\,\vartheta_{11}\left(v+\frac{\omega}{2}\right) = i\varepsilon\,\vartheta_{10}\left(v+\frac{1+\omega}{2}\right),\\
\vartheta_{10}(v) &= \vartheta_{11}\left(v+\frac{1}{2}\right) = \varepsilon\,\vartheta_{00}\left(v+\frac{\omega}{2}\right) = \varepsilon\,\vartheta_{01}\left(v+\frac{1+\omega}{2}\right),\\
\vartheta_{11}(v) &= -\vartheta_{10}\left(v+\frac{1}{2}\right) = -i\varepsilon\,\vartheta_{01}\left(v+\frac{\omega}{2}\right) = -i\varepsilon\,\vartheta_{00}\left(v+\frac{1+\omega}{2}\right),
\end{aligned} \tag{8}$$

where, as an abbreviation,

$$\varepsilon = e^{\frac{\pi i\omega}{4}+\pi i v}$$

is put.

By (3), (4),

$$\vartheta_{00}(-v) = \vartheta_{00}(v), \quad \vartheta_{01}(-v) = \vartheta_{01}(v), \quad \vartheta_{10}(-v) = \vartheta_{10}(v), \quad \vartheta_{11}(-v) = -\vartheta_{11}(v), \tag{9}$$

that is, $\vartheta_{00}(v), \vartheta_{01}(v), \vartheta_{10}(v)$ are even functions, while $\vartheta_{11}(v)$ is an odd function.

By § 17, (12), the four functions (7) vanish respectively for the following values of the argument

$$\frac{1+\omega}{2}, \qquad \frac{\omega}{2}, \qquad \frac{1}{2}, \qquad 0,$$

thus for certain half-periods. Of importance are the values which all the functions (7) assume for these argument values, and these can be reduced by means of table (8) to the three

$$\vartheta_{00}(0) = \vartheta_{00}, \qquad \vartheta_{01}(0) = \vartheta_{01}, \qquad \vartheta_{10}(0) = \vartheta_{10}.$$

Thus from (8) one obtains the following system of formulae:

$$\begin{aligned}
\vartheta_{00} &= \vartheta_{01}\left(\frac{1}{2}\right) = \varepsilon_0\vartheta_{10}\left(\frac{\omega}{2}\right) = \varepsilon_0\vartheta_{11}\left(\frac{1+\omega}{2}\right), \\
\vartheta_{01} &= \vartheta_{00}\left(\frac{1}{2}\right) = -i\varepsilon_0\vartheta_{11}\left(\frac{\omega}{2}\right) = i\varepsilon_0\vartheta_{10}\left(\frac{1+\omega}{2}\right), \\
\vartheta_{10} &= \vartheta_{11}\left(\frac{1}{2}\right) = \varepsilon_0\vartheta_{00}\left(\frac{\omega}{2}\right) = \varepsilon_0\vartheta_{01}\left(\frac{1+\omega}{2}\right),
\end{aligned} \tag{10}$$

where

$$\varepsilon_0 = e^{\frac{\pi i\omega}{4}},$$

and

$$\vartheta_{11}(0) = 0, \quad \vartheta_{10}\left(\frac{1}{2}\right) = 0, \quad \vartheta_{01}\left(\frac{\omega}{2}\right) = 0, \quad \vartheta_{00}\left(\frac{1+\omega}{2}\right) = 0. \tag{11}$$

The squares of the functions (7),

$$\vartheta_{00}^2(v), \quad \vartheta_{01}^2(v), \quad \vartheta_{10}^2(v), \quad \vartheta_{11}^2(v), \tag{12}$$

are Θ_{00}-functions of the second order, and consequently there are two linear relations between them. If we assume these relations in the form

$$\vartheta_{10}^2(v) = A\vartheta_{01}^2(v) + B\vartheta_{11}^2(v), \qquad \vartheta_{00}^2(v) = A'\vartheta_{01}^2(v) + B'\vartheta_{11}^2(v),$$

then the coefficients can be easily determined on the basis of formulae (10), (11), if one sets $v = 0$ and $v = \frac{\omega}{2}$. Thus one obtains

$$\vartheta_{01}^2\vartheta_{10}^2(v) = \vartheta_{10}^2\vartheta_{01}^2(v) - \vartheta_{00}^2\vartheta_{11}^2(v), \qquad \vartheta_{01}^2\vartheta_{00}^2(v) = \vartheta_{00}^2\vartheta_{01}^2(v) - \vartheta_{10}^2\vartheta_{11}^2(v), \tag{13}$$

and from this also, putting $v = \frac{1}{2}$,

$$\vartheta_{00}^4 = \vartheta_{01}^4 + \vartheta_{10}^4. \tag{14}$$

By any two of the squares (12) all functions Θ_{00} of the second order can be represented linearly; but the remaining Θ-functions of the second order can also be composed from the functions ϑ_{g_1,g_2}, for for each characteristic one obtains two linearly independent products, of which one is an even and the other an odd function, namely:

$$\begin{array}{lll}
\vartheta_{00}(v)\vartheta_{01}(v), & \vartheta_{10}(v)\vartheta_{11}(v) & \text{characteristic } (0,1), \\
\vartheta_{00}(v)\vartheta_{10}(v), & \vartheta_{01}(v)\vartheta_{11}(v) & (1,0), \\
\vartheta_{10}(v)\vartheta_{01}(v), & \vartheta_{00}(v)\vartheta_{11}(v) & (1,1).
\end{array} \tag{15}$$

By the same principle all Θ-functions of arbitrary order and arbitrary main characteristic can now be formed from the functions ϑ. To prove this, we denote by Θ_0, Θ_1 any two of the ϑ-squares (12), and by $F^{(n)}(\Theta_0, \Theta_1)$ an entire rational homogeneous function of nth order of the two arguments Θ_0, Θ_1. We obtain hereafter for the Θ-functions of mth order $\Theta^{(m)}(v)$ the following expressions.

For even m:

even functions:

$$\Theta_{00}^{(m)}(v) = F^{(m/2)}(\Theta_0, \Theta_1), \qquad \Theta_{01}^{(m)}(v) = \vartheta_{00}(v)\vartheta_{01}(v)F^{(m/2-1)}(\Theta_0, \Theta_1),$$
$$\Theta_{10}^{(m)}(v) = \vartheta_{00}(v)\vartheta_{10}(v)F^{(m/2-1)}(\Theta_0, \Theta_1), \qquad \Theta_{11}^{(m)}(v) = \vartheta_{10}(v)\vartheta_{01}(v)F^{(m/2-1)}(\Theta_0, \Theta_1),$$

odd functions:

$$\Theta_{00}^{(m)}(v) = \vartheta_{00}(v)\vartheta_{10}(v)\vartheta_{01}(v)\vartheta_{11}(v)F^{((m-4)/2)}(\Theta_0, \Theta_1),$$
$$\Theta_{01}^{(m)}(v) = \vartheta_{10}(v)\vartheta_{11}(v)F^{(m/2-1)}(\Theta_0, \Theta_1), \qquad \Theta_{10}^{(m)}(v) = \vartheta_{01}(v)\vartheta_{11}(v)F^{(m/2-1)}(\Theta_0, \Theta_1),$$
$$\Theta_{11}^{(m)}(v) = \vartheta_{00}(v)\vartheta_{11}(v)F^{(m/2-1)}(\Theta_0, \Theta_1). \tag{16}$$

For odd m:

even functions:

$$\Theta_{00}^{(m)}(v) = \vartheta_{00}(v)F^{((m-1)/2)}(\Theta_0, \Theta_1), \qquad \Theta_{01}^{(m)}(v) = \vartheta_{01}(v)F^{((m-1)/2)}(\Theta_0, \Theta_1),$$
$$\Theta_{10}^{(m)}(v) = \vartheta_{10}(v)F^{((m-1)/2)}(\Theta_0, \Theta_1), \qquad \Theta_{11}^{(m)}(v) = \vartheta_{00}(v)\vartheta_{01}(v)\vartheta_{10}(v)F^{((m-3)/2)}(\Theta_0, \Theta_1),$$

odd functions:

$$\Theta_{00}^{(m)}(v) = \vartheta_{01}(v)\vartheta_{10}(v)\vartheta_{11}(v)F^{((m-3)/2)}(\Theta_0, \Theta_1),$$
$$\Theta_{01}^{(m)}(v) = \vartheta_{00}(v)\vartheta_{10}(v)\vartheta_{11}(v)F^{((m-3)/2)}(\Theta_0, \Theta_1),$$
$$\Theta_{10}^{(m)}(v) = \vartheta_{00}(v)\vartheta_{01}(v)\vartheta_{11}(v)F^{((m-3)/2)}(\Theta_0, \Theta_1), \quad \Theta_{11}^{(m)}(v) = \vartheta_{11}(v)F^{((m-1)/2)}(\Theta_0, \Theta_1). \tag{17}$$

That all Θ-functions are representable in this form follows, on the basis of § 20, from the following three considerations.

1. Between even and odd functions no linear dependence can exist unless a linear dependence already exists between the even functions alone or between the odd functions alone.

2. Since two of the ϑ-squares (12) do not stand in a constant ratio, there also exists no equation of the form

$$F^{(n)}(\Theta_0, \Theta_1) = 0.$$

3. The total number of constants which occur, according to (16), (17), in the even and odd functions belonging to one and the same characteristic is exactly equal to the order m.